

CAD FIT CHECK - will the real parts fit the printed parts?

[!] SUPERSEDED IN PART - read CAD_CHANGELOG_v7.5.md alongside this This document audited **v7.2/v7.3**. On 2026-07-30 a repair pass (**v7.5**) fixed every defect that did not require calipers. The findings below are preserved as the *record of what was wrong*, but the following are **FIXED and no longer describe the source**: | Section | Defect | Status |
|---|---|---| | §2a | Hall pocket entirely inside the cam pocket | [OK] rebuilt on the plate underside at (0, 17.35) | | §2c | Homing magnet craters the cam tracks | [OK] blind 1.2mm pocket, 3x1mm magnet (it craters **three** tracks, not two - §2c undercounts) | | §3a | Right motor screw opens into the spring cavity | [OK] spring cavity deleted; pilots instead of clearance holes | | §3b | Left ear hole 0.35mm from the plate edge | [OK] base_length 56 -> 58, wall now 1.35mm | | §5 | Muscle-board bosses clash with the ULN2003 | [OK] bosses now default OFF | | §6a | ESP32 overruns the pod by 1.75mm | [OK] pod_length 64 -> 68 | | §6c | USB cutout ~2mm too high | [OK] arithmetic bug fixed (ignored hdr_channel_depth) | | §6d | USB cutout too small for a real plug | [OK] 10x7 -> 13x9 | | §7a | Jack cradle overruns the cavity | [OK] fixed, plus an assert so it can't recur silently | **Still open after the completed measurements** (see MEASUREMENTS_NEEDED.md): §4c motor/cam vertical-stack re-derivation - §2b physical Hall-pocket dry-fit - §7b selecting the actual threaded panel-mount jack.

2026-07-31 measurement-research update The online and owned-part pass is recorded in MEASUREMENT_RESEARCH.md. - Motor M1-M9 now include the fit-critical owned readings; M5 is 3.0mm across flats. - M11b is measured at 1.6mm. The recess is also 1.6mm, so geometry passes but a printed dry-fit is still required because there is no manufacturing margin. - M18 is a 30-pin USB-C ESP32 and M21 is measured at 25.6mm. The live socket channels now use it. - The owned power part is photo-identified as an inline female DC pigtail jack with red/black wires. It has no thread/nut; keep it for testing and select a nut-mounted 5.5x2.1mm jack for the final pod.

Date: 2026-07-29 **Scope:** physical fit of purchased/owned components inside the v7.2 printed parts. **Status of the build:** nothing has ever been assembled. This is the last check before spending.

How to read this document

Every statement is tagged:

- **[SCAD]** - read directly out of the OpenSCAD source in cad/scad/. This is fact about the *printed part*.
- **[SPEC]** - a published nominal figure for a standard catalogue part (28BYJ-48, 6x6 tactile switch, 5.5/2.1 barrel jack, micro-USB). Standard parts still vary between clone vendors.
- **[ASSUMED]** - a number somebody typed into the CAD about a physical object that has never been measured. Treat as fiction until a caliper says otherwise.

Historical note: when this v7.2/v7.3 audit was written, the "Real part needs" column was unmeasured. Several owned-part dimensions have since been measured; use MEASUREMENT_RESEARCH.md as the current evidence table.

1. SUMMARY TABLE

Component	CAD provides [SCAD]	Real part needs	Verdict	Action
Hall sensor (MH-Sensor-Series / KY-024 module)	5.3 x 4.3 x 3.0 mm pocket at (0, +20) in base_plate	PCB module ~15 x 11 x 1.6 mm + ~11 mm of 4-pin header [ASSUMED]	FAILS	Desolder the bare sensor, or move the module out of the plate entirely
Hall pocket itself	pocket floor z=2; dia 46 cam pocket floor also z=2, and the pocket lies entirely inside r=23	any pocket at all	FAILS	The pocket is 100% subsumed by the cam pocket - it does not exist as a feature
Cam homing magnet dia 3x2	dia 3.2 x 2.0 pocket at r=17.35 in a 2.0 mm disc floor	3x2 mm disc magnet - not owned	FAILS	Pocket is a through-hole that punches a dia 3.2 crater through cam tracks 2 and 3
28BYJ-48 motor body	dia 29 seat, 35 mm hole spacing, dia 4.3 holes, body centre at x=-8, 19 mm of z-space	body dia 28, 35 mm spacing, dia 4.2, ~8 mm offset, 19 mm can [SPEC]	UNVERIFIED	Never measured. 8 caliper readings required (§4)
Motor right mounting screw	hole at (+9.5, 0) falls inside the 22 x 16 spring-cavity through-window	material to clamp against	FAILS	Motor can only be held by ONE screw

Component	CAD provides [SCAD]	Real part needs	Verdict	Action
Motor screw head vs cam	left hole at x=-25.5 is clear; right hole at x=+9.5 is under the dia 46 cam disc	screw head clearance	FAILS	No counterbore modelled; head fouls the cam floor
Cam on shaft (double-D bore)	dia 5.2 clipped to 3.2 flats; through-hole , real engagement = hub 4 + floor 2 = 6 mm	5.0 mm shaft, 3.0 mm across flats, ~10 mm long [SPEC]	UNVERIFIED / marginal	Clearances fine on paper; z-stack is wrong (\$5)
Cam z-stack vs motor face	hub bottom z=39, motor top face z=41	hub cannot go below the motor face	FAILS	Cam bottoms on the motor and sits ~2 mm high -> all six dots permanently proud
Shaft protrusion	cam surface z=45; a 10 mm shaft reaches z=51	shaft must not enter the dot cluster	FAILS (conditional)	Shaft passes through the open bore into the linkage risers at x=+/-2.4
ULN2003 module, wires soldered flat	36 x 46 x 16 pocket + 2 mm mid-plate relief slot = 18 mm	~35 x 32 x ~12 mm [ASSUMED, established earlier]	FITS	Solder wires flat. Watch the 0.5 mm/side X clearance
ULN2003 module, vertical Dupont jumpers	same 18 mm max	~20 mm as-wired [ASSUMED]	FAILS	Confirmed: 20 > 18. No vertical jumpers
ULN2003 vs muscle-board bosses	4 mm tall bosses at (+/-14, +/-19) rise from the pocket floor	flat floor	UNVERIFIED	If the board is ~35 mm in Y it clips the y=+/-19 bosses
ESP32 DevKit length in pod	cavity +/-28 in X; board 51.5 centred at x=+4 -> spans -21.75 .. +29.75	board must end inside x=+28	FAILS	1.75 mm interference with the +X (dock) wall
ESP32 DevKit width / row pitch	devkit_width 28.0, hdr_row_pitch 25.4 [ASSUMED]	30-pin DOIT V1 is commonly ~25.4 wide / 0.9" pitch; 38-pin is ~28 / 1.0" [SPEC-ish]	UNVERIFIED	The CAD numbers describe the 38-pin board; the docs say 30-pin
USB cutout alignment	cutout z 12.5 .. 19.5; board top sits at z=10.5 (8.5 strip ? 1.0 channel recess)	port body ~10.5 .. 13.0	FAILS	Hole sits ~2 mm too high; only ~0.5 mm overlaps the port
USB cutout size	10 x 7 mm	micro-USB overmould commonly ~12 x 8 [SPEC-ish]	UNVERIFIED	Plug may not enter
Headroom under pod lid	board top z~10.5, cavity top z=54	a few mm	FITS	43 mm spare - grossly over-tall, but no clash
Barrel jack hole	dia 11.5 at (-22, +18) on the lid	5.5/2.1 panel jack, ~dia 11 thread [SPEC]	UNVERIFIED	Owned part is an inline pigtail, not the modelled panel jack
Barrel jack cradle	inner 12 x 9.5 x 11, walls 1.5 -> outer reaches x = -29.5	must fit inside cavity x >= -28	FAILS	1.5 mm interference - the lid will not close. And all three dims are PLACEHOLDERS
Docking magnets	dia 8.4 x 1.2 pockets, 2 per face at y=+/-14, 4 mm wall	8 x 1 mm NdFeB discs - measured, in hand	FITS	Only re-confirm the newly bought batch is the same 8x1
Nav buttons - cap stem	dia 3.8 stem projects 4.5 mm behind the flange through a dia 4.2 hole	must cross a 4 mm wall	FITS (nominal)	Stem reaches 0.5 mm inside; 0.4 mm diametral clearance allows resin-in-PETG sliding
Nav buttons - switch reach	The 6 mm switch body is stopped by the pod's inner front wall; cage back wall takes press load	plunger must meet the stem	FITS (nominal)	The front wall is the forward datum; bench-test the actual switch's plunger before final resin batch
6x6x5 tactile switch in cage	6.4 mm slot, 7.2 mm depth, legs vertical	6.0 x 6.0 x 5.0 body [SPEC]	FITS	Body and revised cap reach fit on the nominal switch geometry

Component	CAD provides [SCAD]	Real part needs	Verdict	Action
Dot insert into top plate	body 11.0 into 11.2 opening; flange 15.0 into 15.2 rebate; heights 2.0 / 1.2; glue shelf 2.0 mm wide, 105.6 mm ²	resin tile printed from the same file	FITS	Verified numerically. Only risk is FDM square holes printing undersize
Base-plate standoffs	dia 6 posts at x=+-26, plate half-length 28 -> 1 mm hangs off the edge	supported footprint	UNVERIFIED / cosmetic	Prints as a small unsupported overhang; not structural

Required outer-box changes - handout checklist

These items are deliberately **not** applied to `outer_box.scad` yet. They either require the real motor dimensions or a hardware-choice decision; changing them from catalogue numbers would simply move the fit failure somewhere else.

Priority	Required change	Why	Prerequisite
Blocker	Re-derive <code>base_plate_z</code> and the motor-height allowance	The current 41 mm motor-face height is an unmeasured assumption. It controls the cam/linkage vertical stack.	Measure the actual motor can height and shaft height.
Blocker	Redesign the base-plate spring cavity around both motor ears	The right motor screw opens into the existing 22 x 16 mm through-window, leaving a one-screw mount.	Measure the actual motor mounting-ear envelope and hole centres.
Blocker	Widen/reposition the base-plate support footprint if the motor-ear fix needs it	The left ear currently has only a 0.35 mm edge wall and the base standoffs overhang the plate by 1 mm.	Motor-ear measurement, then confirm the revised plate still clears the 60 mm shell cavity and corner bosses.
Required before installing the owned Hall module	Add a real Hall-module bracket in the outer-box electronics area, or commit to a measured bare-sensor design	The current base-plate pocket is entirely subsumed by the cam pocket and cannot hold the owned module.	Decide module versus desoldered sensor; measure its footprint and sensing-face offset.
Decision required	Remove/relocate the four future-muscle-board bosses when using the ULN2003 module	A 35 mm-wide ULN2003 can collide with the y=+-19 mm bosses in the outer-box electronics pocket.	Measure the actual ULN2003 board and choose prototype driver versus future muscle PCB.

Do not alter the 68 x 68 mm shell, docking windows, magnet pockets, or 4 mm walls as part of this repair.

Those features are independent of the motor and have already been physically fit-tested. The next safe sequence is: measure hardware -> update the dependent base/mid/cam/ outer-box dimensions together -> render the full assembly -> print a cheap proving part.

2. HALL SENSOR - FAILS, twice over

2a. The pocket does not physically exist

[SCAD] `base_plate.scad`:

- `hall_pocket_h` = 3, cut from `base_thickness` - `hall_pocket_h` = **z = 2** upward.
- `cam_pocket_depth` = 3, `cam_pocket_diameter` = 46, cut from **z = 2** upward, radius 23.
- Hall pocket footprint: x ? [-2.65, +2.65], y ? [17.85, 22.15]. Its farthest corner is at

radius ?(2.652 + 22.152) = **22.31 mm < 23 mm**.

The hall pocket lies **entirely inside the cam pocket, at exactly the same depth**. Subtracting it removes nothing. There is no rectangular recess on the printed part - just flat cam-pocket floor.

Worse: the cam disc's underside *sits on that floor* (`cam_flat_z` = 45, disc floor 2 mm, pocket floor `z`=43 in box coordinates). There is **zero vertical space** above the "pocket" for any sensor, bare TO-92 or otherwise. The wire channel (`hall_wire_w`=2 x `hall_wire_d`=1) does survive, but only in the strip of plate outside `r`=23.

2b. The module is far too big anyway

[ASSUMED - MEASURE THIS] The blue MH-Sensor-Series / KY-024-style board Mridul owns (identified by Codex on 2026-06-03 as an analog+digital module with AO/DO/GND/VCC) is a PCB in the region of **15 x 11 mm, ~1.6 mm**

thick, plus a 4-pin 2.54 mm header adding roughly 8-11 mm of height. Those figures are typical for that family of modules; they are **not** measured from his part.

Against a 5.3 x 4.3 x 3.0 pocket sized for a bare SS49E/A3144 TO-92 body:

- length: needs ~15, has 5.3 -> **~10 mm short**
- width: needs ~11, has 4.3 -> **~7 mm short**
- height: needs ~10-13 with the header, has 3 (and really has 0, see 2a) -> **hopeless**

Fixes, in order of preference

1 Desolder the bare sensor from the module (recommended). The MH-series board carries a standard TO-92 hall device plus an LM393 comparator, an LED and a trimpot. Unsolder the three legs, mount the bare TO-92 in the plate, and run three wires back to the module (or straight to the ESP32 with a pull-up, since the breadboard test already ran the analog output at 3.3 V). This keeps the CAD footprint honest - but §2a still has to be fixed first, because the pocket currently isn't there.

1 Move the module off the base plate entirely. Mount it on the electronics-pocket floor or on a wall of the outer box, with the sensing face pointing up at the cam, and run wires. The cam magnet is 3 mm dia - flux falls off fast, so this only works if the module can get within a few mm of the magnet path. This is the lowest-CAD-risk option but needs a new bracket.

1 Enlarge and deepen the pocket. Requires all three of: (a) make `hall_pocket_h` > 3 so the pocket floor drops below the cam pocket floor (e.g. 4.0-4.5 mm, leaving 0.5-1.0 mm of plate underneath); (b) grow `hall_pocket_w/d` to the measured module footprint + 0.4 mm; (c) move `hall_pocket_y` outboard past `y=23` so it is no longer under the disc, **or** add a header escape slot. Note the plate is only 5 mm thick and the +Y edge is at `y=25` - there is very little room. Option 1 or 2 is more realistic.

2c. BONUS DEFECT FOUND - the cam homing-magnet pocket is a hole through two tracks

[SCAD] `braille_cam.scad`: `magnet_dia=3.0, magnet_depth=2.0`, cut as `cylinder(d=3.2, h=3)` from `z=-1`, i.e. **`z = -1 .. +2`**, at `r = 17.35`, angle 90deg. The disc floor is `disk_base_thickness = 2.0`, spanning `z = 0 .. 2`. The cut therefore removes the **entire floor thickness** - it is a through-hole, not a pocket ("leaves 0mm floor" is even in the comment). And at `r = 17.35 +/- 1.6` it spans `r = 15.75 .. 18.95`, which overlaps **track 2** (`r 15.4-17.0`, drives dot 1) and **track 3** (`r 17.1-18.7`, drives dot 2).

Where those tracks are in their DOWN state the cam surface at that angle is simply **missing** - a dia 3.2 mm hole in a running surface that a 1.0 mm-thick linkage foot crosses once per revolution. The foot will drop in and jam.

Fix: shrink `magnet_depth` to ~1.4 mm and thicken the local floor, or (better) move `magnet_radius` inboard to `r < 12` (inside `inner_radius`, under no track at all) and move the hall sensor to match. Moving the magnet inboard also weakens coupling, so re-check the sensor gap. Also: **the 3x2 mm homing magnet is not in hand** (docs/SOURCING.md §2) - it still has to be bought.

3. MOTOR - NEVER MEASURED. Treat every motor number as fiction.

[!] **THE 28BYJ-48 IN THIS PROJECT HAS NEVER BEEN TOUCHED WITH A CALIPER.** `base_plate.scad`, `mid_plate.scad` and `braille_cam.scad` all encode motor dimensions that were typed from datasheets and forum posts. Mridul has photographed a motor whose shaft looked different from the model. Every verdict in this section is conditional on measurement.

[SCAD] what the CAD assumes:

Parameter	Value	File
body diameter	dia 29 (seat pocket, 1.5 mm deep)	<code>base_plate.scad</code>
retaining collar ID	dia 29.5, 8 mm tall, at <code>x=-8</code>	<code>mid_plate.scad</code>
mount-hole spacing	35 mm	<code>base_plate.scad</code>
mount-hole diameter	dia 4.3	<code>base_plate.scad</code>
shaft offset from body centre	-8 mm (shaft at <code>x=0</code> , body at <code>x=-8</code>)	<code>base_plate.scad</code> , <code>mid_plate.scad</code>
can height allowance	19 mm (<code>z=22 .. 41</code>)	<code>outer_box.scad</code> <code>base_plate_z</code>
shaft	dia 5.0, 3.0 across flats, 10 mm long	<code>braille_cam.scad</code> comments

[SPEC] The published nominal 28BYJ-48 is: can dia 28 x 19 mm, mounting bracket holes 35 mm apart at dia 4.2, shaft offset ~8 mm from the can centre, shaft dia 5.0 with two flats ~3.0 mm apart, protruding ~10 mm. So the CAD is *plausible*. Clone variance and the 19 mm z-budget with **zero** slack are the risks.

Two hard failures that exist regardless of measurement

3a. The right-hand motor screw has nothing to screw into. [SCAD] `spring_cavity()` is `cube([22, 16, 20], center=true)` - a through-window spanning x ? [-11, +11], y ? [-8, +8]. The right mounting hole is at x = -8 + 35/2 = **+9.5**, y = 0, dia 4.3 -> spans x ? [7.35, 11.65]. That is inside the window. There is no plate material there. **The motor can only be bolted by its left ear (x=-25.5)**. A single-screw stepper will rock and lose steps.

3b. The left ear hole is 0.35 mm from the plate edge. x = -25.5, dia 4.3 -> outer edge at -27.65; plate half-length is 28. A 0.35 mm wall on a 0.4 mm nozzle is not a wall. The screw will break out.

Fix for both: shrink/reshape `spring_cavity` so it clears x=+9.5 (e.g. two smaller windows, or move it to -Y), and widen `base_length` from 56 to at least 60 (the box cavity is 60 - so this also needs the cavity or the standoff pattern revisited).

Measurements required (feeds -> CAD parameter)

- 1 Can outer diameter -> `motor_body_diameter` (`base_plate`), collar ID (`mid_plate`)
 - 2 Can height, bracket face to bottom -> `base_plate_z` in `outer_box.scad`
 - 3 Shaft centre to can centre -> `motor_body_x_offset`, `motor_x_offset`
 - 4 Shaft diameter -> cam bore dia (currently 5.2)
 - 5 Shaft flat-to-flat -> cam bore flat clip (currently 3.2)
 - 6 Shaft length above the bracket face -> `cam_hub_h` + `shaft_bore_depth` + z-stack
 - 7 Mount-hole centre-to-centre -> `motor_mount_spacing`
 - 8 Mount-hole diameter -> `motor_mount_hole`
 - 9 Shaft boss (raised collar around the shaft) diameter and height -> cam hub OD (currently dia 9)
-

4. CAM ON SHAFT - clearances OK on paper, z-stack is wrong

4a. The bore profile

[SCAD] `braille_cam.scad: intersection(cylinder(r=2.6), cube([10, 3.2, ...]))` -> a dia 5.2 cylinder clipped to 3.2 mm across the flats, on both sides (true double-D).

[SPEC] Nominal 28BYJ-48 shaft: **dia 5.0 mm, 3.0 mm across the flats, ~10 mm long.**

- radial clearance: $(5.2 - 5.0)/2 = 0.10$ mm per side
- flat clearance: $(3.2 - 3.0)/2 = 0.10$ mm per side

That is correct in intent but *very* tight for a printed part. On resin the bore will come out slightly undersize and will need reaming/filing; on FDM it will be far undersize. Plan on hand-fitting. Do **not** loosen it blindly - a sloppy double-D on a 5.625deg indexing cam translates directly into dot-position error.

4b. Engagement depth is 6 mm, not the 8 mm the comment claims

[SCAD] The bore is subtracted over local z = 75 .. +4 (`translate([0, 0, -hub_h-1])`), h = `shaft_bore_depth` + 1 = 9). Material exists only over z = 74 .. +2 (hub 4 mm + disc floor 2 mm). So:

- **actual maximum engagement = 6.0 mm**, and
- the bore is a **through-hole** - it opens onto the top of the disc floor. There is no ceiling and

therefore **no axial stop** locating the cam on the shaft.

The header comment "Hub 4mm below disc + bore 4mm into disc floor = 8mm total engagement" is stale: the disc floor is 2 mm, not 4 mm.

6 mm of double-D on a ~10 mm shaft is *acceptable* engagement for this torque, but the missing axial stop means cam height depends entirely on how far it is pushed on during assembly - with `pin_lift` = 0.8 mm and dot flushness already fought over in v7.2, that is a real risk. Add a 0.5-1.0 mm shoulder at the top of the bore, or a printed depth gauge.

4c. The vertical stack does not close - SHOWSTOPPER (conditional on motor measurement)

Working in outer-box world z [SCAD]:

feature	z
motor top / bracket face = base-plate underside	41
base plate top	46
cam pocket floor (base_thickness ? cam_pocket_depth)	43
cam disc floor bottom -> top (cam_flat_z)	43 -> 45
cam hub bottom (4 mm below the disc)	39
shaft top, if the shaft is 10 mm [SPEC]	51

Two consequences:

- **The hub is 2 mm too long.** Only 2 mm of space exists between the cam pocket floor (43) and the motor face (41), but hub_h = 4. The hub bottoms out on the motor and the disc sits **2 mm proud of its pocket**, pushing cam_flat_z to 47. Since link_total_h is derived from cam_flat_z = 45, **every one of the six dots ends up ~2 mm permanently proud** - exactly the class of bug v7.2 just fixed for a 0.8 mm offset.

- **The shaft sticks out the top.** With the bore open, a 10 mm shaft reaches z=51, i.e. **6 mm above the cam surface (**4 mm above it even in the mis-seated position**). **The dot cluster sits at x = +/-2.4, y = 0/+/-2.6 - directly over a dia 5 shaft (r=2.5). The linkage risers occupy 3.5 to 12.2 mm above the cam surface at exactly those coordinates.** The shaft will foul the dot-2 and dot-5 risers.**

Fixes (all need measurement 6 above first): shorten hub_h to the real gap; counterbore the base plate around the shaft so the hub can drop; trim the motor shaft; or lower the motor by increasing the z-budget. Do not guess - measure the shaft, then re-derive the whole stack in one pass.

5. ULN2003 IN THE 36 x 46 x 16 POCKET - confirmed

[SCAD] outer_box.scad: electronics pocket is cube([36, 46, 16]) centred at x=0,y=0, floor at z=4, top at z=20 (X=36, Y=46). mid_plate.scad adds a through relief slot translate([3, -23, -1]) cube([17, 8, 4]) - 17 mm (x=3..20) by 8 mm (y=-23..-15) - giving **2 mm extra headroom** at that corner only.

configuration	height [ASSUMED, established earlier]	available	verdict
vertical Dupont jumpers plugged into the ULN2003 header	~20 mm	16 mm (18 mm at the relief slot)	FAILS by ~2-4 mm
wires soldered flat to the board	~12 mm	16 mm	FITS, 4 mm spare

Confirmed: solder the wires flat. No vertical jumpers, no exceptions. This already matches docs/WIRING_AND_ASSEMBLY.md.

Three secondary cautions:

1 Plan fit is tight in X. A ULN2003 driver board is commonly ~35 x 32 mm [ASSUMED - measure].

If its 35 mm side runs along X it has **0.5 mm per side** in a 36 mm pocket. Orient the 35 mm side along Y (46 mm) instead.

1 Muscle-board bosses are in the way. Four dia 4 x 4 mm tall bosses stand on the pocket floor at (+/-14, +/-19). If the ULN2003 is ~35 mm in Y (+/-17.5) it clips the y=+/-19 bosses. The pocket is currently spec'd for *two different boards* (the future 34x44 muscle PCB and the off-the-shelf ULN2003) and only one of them can be there at a time - decide which.

1 Floor wire gutters are cut 2.5 mm deep at x ? [-18,-11] and [11,18] across the whole pocket.

A board laid on the floor will bridge them; fine, but it will not sit flat if it has through-hole tails on the underside.

The mid-plate relief slot is **correctly placed** for a JST/header edge pointed at the (+X, ?Y) corner - it clears the motor collar (whose max x at y=?15 is +0.5) and the boss hole at (26, ?21). Orient the module accordingly.

6. ESP32 DEVKIT IN THE POD - FAILS on length and on USB alignment

[SCAD] esp32_pod_params.scad / esp32_pod_shell.scad: pod_int_length = 64 ? 2-4 = 56 -> cavity spans x ? [28, +28]. devkit_length = 51.5, devkit_x_offset = +4.

6a. Length - 1.75 mm interference

Board spans $x = 4 \pm 25.75 = \pm 21.75$ to $4 + 25.75 = +29.75$. Cavity ends at **+28**. **The board overruns the +X (dock) wall by 1.75 mm.** It will not go in.

The header channels themselves are fine (they span $x = \pm 16 \dots \pm 24$, retention walls to ± 25.5 , all inside the cavity) - it is the *board overhanging the sockets* that hits the wall. The antenna grille recess in that wall only starts at $z=23$, far above the board at $z \sim 10.5$, so it does not help.

Fix: reduce `devkit_x_offset` to $\sim +2$ (giving ~ 0.25 mm at each end - still too tight) or lengthen the pod. Realistically `pod_length` needs to go from 64 to ~ 68 , which conveniently also matches the cell footprint. **Measure the board first.**

6b. Width and header pitch - M21 resolved at 25.6mm

[SCAD/current] `devkit_width = 28.0`, `hdr_row_pitch = 25.6`. The owned board is confirmed as 15 pins per row / 30 total, USB-C, with 25.6mm measured between the row centres.

[SPEC, still verify owned board] Pin count does not identify the transverse footprint. Commercial 30-pin carriers explicitly support 0.9, 1.0, and 1.1 inch row spacing - 22.86, 25.4, and 27.94 mm. Matching USB-C CH340C listings are approximately 51.5 x 28.5 mm overall.

The live 25.6mm channel spacing now follows the owned board rather than the nominal 25.4mm family. This resolves the header-location blocker; board insertion still needs a first-print dry-fit.

6c. USB cutout - ~ 2 mm too high

[SCAD] `usb_z = pod_floor + hdr_strip_h + 1 = 3 + 8.5 + 1 = 12.5`, cutout 10 wide x 7 tall -> spans $z = 12.5 \dots 19.5$.

But `hdr_channel_depth = 1.0` recesses the strips 1 mm into the floor, so the strip top - and therefore the **board underside** - sits at $3 + 8.5 \pm 1 = 10.5$, not 11.5. A micro-USB socket sits on the board and stands ~ 2.5 mm tall -> it occupies $z \sim 10.5 \dots 13.0$.

Overlap with the cutout: $z = 12.5 \dots 13.0$, i.e. **0.5 mm**. The port is essentially behind solid wall. `usb_z` should be ~ 10.0 .

6d. USB cutout size and plug reach

- [SCAD] cutout is 10 x 7 mm in a 4 mm wall.
- [SPEC-ish] a micro-USB plug's *metal shell* is $\sim 7 \times 2.5$ mm, but the **overmoulded body** on a typical cable is $\sim 11\text{--}13$ mm wide and $\sim 7\text{--}9$ mm tall. It will very likely not pass a 10 x 7 hole.
- The board's $\pm X$ edge sits at $x = \pm 21.75$ while the wall inner face is at ± 28 , so the plug must reach **6.25 mm inboard** past the wall to seat. Combined with the overmould, this is unlikely to work. Enlarge to $\sim 13 \times 9$ and reduce the standoff distance.

Current source (v7.7): resolved provisionally. The board was moved toward the USB wall, reducing plug reach to 2.25 mm, and the service opening is now **14 x 9 mm**. This deliberately preserves the 9 mm overmould height clearance while adding 1 mm of width. Measure M23 only if the intended cable body exceeds that envelope.

6e. Headroom under the lid - fine

Cavity top at $z = 54$ (`shell_h = 58 \pm 4`). Board top ~ 10.5 , tallest DevKit component maybe 5 mm -> $z \sim 15.5$. **~ 38 mm of unused headroom.** No clash with the jack cradle either: the cradle bottom sits at $z = 54 \pm 11 \pm 1.5 = 41.5$, and it is at $y = +18$ while the board spans $y = \pm 14$.

The pod is enormously over-tall for its contents. Not a fit failure, but worth knowing before printing 58 mm of PETG.

7. BARREL JACK - UNVERIFIED, and the placeholder cradle already collides

[SCAD] `esp32_pod_lid.scad`:

```
jack_body_w = 9.5;    // PLACEHOLDER
jack_body_l = 12;     // PLACEHOLDER
jack_body_h = 11;     // PLACEHOLDER
cradle_t     = 1.5;
barrel_jack_dia = 11.5; at (x=-22, y=+18)
```

The author has flagged these as placeholders. Two things to know:

7a. Even the placeholder cradle does not fit the pod. Cradle outer extent in $X = \text{barrel_jack_x} \pm \text{jack_body_l}/2 \pm \text{cradle_t} = \pm 22 \pm 6 \pm 1.5 = \pm 29.5$. The pod cavity ends at $x = \pm 28$. **1.5 mm of interference -**

the lid physically cannot close. Since a real screw-terminal jack is almost certainly *bigger* than the 12 mm placeholder, this gets worse, not better. Either move barrel_jack_x inboard (to about ?20) or shrink the cradle.

7b. The modelled jack is the wrong type. dia 11.5 is sized for a **threaded panel-mount 5.5/2.1 jack with a nut** [SPEC]. The owned part is now photo-identified as an **inline female DC pigtail socket** with red/black wires and no panel thread. A round hole gives it nothing to clamp to. Keep it for testing and choose a real nut-mounted jack for the final pod rather than building a bulky cable-body cradle.

Also still open from the 2026-06-03 breadboard review: **the jack's polarity has never been proven.** Check it with a multimeter before applying adapter power.

Before finalizing this feature: select the exact threaded panel-mount jack and use its drawing or measured thread/body dimensions. Do not dimension the final lid around the testing pigtail.

8. MAGNETS - the only genuinely measured component. Re-confirm the new batch.

[SCAD] outer_box.scad and esp32_pod_params.scad both use mag_dia = 8.4, mag_depth = 1.2, teardrop profile, 2 per face at y = +/-14, z = 29, in a 4 mm wall (2.8 mm of material left behind each pocket).

[MEASURED, v6.1] docs/SOURCING.md records "NeFeB disc magnets **8mm dia x 1mm thick** - plenty" and the v6.1 CAD notes say they were measured off the real parts. So 8.4 x 1.2 gives 0.2 mm radial and 0.2 mm axial glue clearance. **FITS.**

Caveats:

- The bill line "10x magnets + bearing" does **not** identify a size. Confirm the batch in hand is the same 8 x 1 discs that were measured in v6.1, not a different order.
- Quantity: one cell needs **4** (2 per face x +/-X faces); one pod needs **2**. So a cell+pod pair needs 6, and any second cell needs another 4. 10 covers cell + pod + spares, but not a second cell.
- The pockets open onto the **outer** face, so each magnet sits flush with the outside with 2.8 mm of wall behind it. That is the right way round for holding force. A 1 mm-thick 8 mm disc is still a weak magnet - two per face is not much dock retention, so expect the chain to be light-duty.
- The "bearing" on the bill is **no longer needed** - v7.1 replaced the bearing-ball braille dot with a printed 1.5 mm dome (docs/SOURCING.md §3).
- **Separate gap:** the **cam homing magnet (3 x 2 mm)** is a different part and is **not in hand**.

See §2c - and note its pocket is currently broken.

9. DOT INSERT INTO TOP PLATE - FITS (verified numerically)

[SCAD] From mech_layout.scad / dot_insert.scad / top_plate.scad:

feature	insert	plate pocket	clearance
body (square)	insert_body = 11.0	insert_body + insert_fit = 11.2	0.1 mm/side
flange (square)	insert_flange = 15.0	insert_flange + insert_fit = 15.2	0.1 mm/side
body height	insert_h ? insert_flange_h = 2.0	opening runs full 4 mm thickness	ok
flange height	insert_flange_h = 1.2	rebate from z=2.0 to 3.2 = 1.2	exact
total height	insert_h = 3.2	plate_thickness ? finger_pad_depth = 4.0 ? 0.8 = 3.2	flush

- **Glue shelf:** rebate floor at z = 2.0, ring width $(15.2 - 11.2)/2 = 2.0$ mm all round, area = $15.22 - 11.22 = 105.6$ mm². Matches the documented ~106 mm².
- **Spring recess ceiling:** spring_bores cut z = ?0.01 .. 2.0, i.e. exactly the body height, so

the ceiling is the flange underside at insert-z 2.0 = linkage-y 11.0. This matches the mech_layout .scad comment "dot DOWN 8.0 -> 11.0 = 3.0 mm". **Consistent.**

- **Dot clearance:** plate_hole_dia 1.7 vs dot_dome_dia 1.5 -> 0.1 mm/side. Correct.
- **Feature envelope:** dots span x = +/-2.4, y = +/-2.6; plus the dia 2.2 spring bore radius that is x = +/-3.5, y = +/-3.7 - comfortably inside the 11 mm body.

Verdict: FITS. The only real risk is manufacturing, not geometry:

1 The plate is PETG on a 0.4 mm nozzle. A square 11.2 mm hole prints undersize and picks up elephant's foot on the first layers. Expect to **file the opening** or add a 0.15-0.2 mm chamfer.

1 0.1 mm/side is a *press* fit, not a *glue* fit. Thin CA will wick in; epoxy will not - it needs ~0.15-0.2 mm. Use thin CA on the shelf, per the assembly note in dot_insert .scad.

1 Wipe squeeze-out off the top face before it cures or the domes jam. Already documented.

10. MEASUREMENTS REQUIRED BEFORE ASSEMBLY

Every one of these must be taken with calipers on **the actual part in hand**, not from a datasheet or a product listing. The right-hand column is the CAD symbol it feeds.

Motor - 28BYJ-48 (blocks base_plate, mid_plate, braille_cam, outer_box)

#	Measure	Feeds
1	Can outer diameter	motor_body_diameter (base_plate); collar ID 29.5 (mid_plate)
2	Can height, bottom face to bracket face	base_plate_z = 41 (outer_box)
3	Shaft centre -> can centre distance	motor_body_x_offset = ?8; motor_x_offset = ?8
4	Shaft diameter	cam bore cylinder dia 5.2 (r=2.6, braille_cam)
5	Shaft flat-to-flat width	cam bore clip cube([10, 3.2, ...])
6	Shaft length above the bracket face	hub_h = 4, shaft_bore_depth = 8, and the whole z-stack
7	Shaft boss (raised collar) diameter and height	cam hub OD dia 9 (r=4.5)
8	Mount-hole centre-to-centre	motor_mount_spacing = 35
9	Mount-hole diameter	motor_mount_hole = 4.3
10	Bracket ear outer width / thickness	base_length = 56 (the 0.35 mm edge wall problem)

Hall sensor (blocks base_plate)

#	Measure	Feeds
11	MH-Sensor module PCB length x width x thickness	hall_pocket_w = 5.3, hall_pocket_d = 4.3 - or the decision to desolder
12	Header height above the PCB (or below, if it must lie flat)	hall_pocket_h = 3
13	If desoldering: bare TO-92 body width x depth x height, and lead pitch	hall_pocket_w/d/h, hall_wire_w/d

Barrel jack (blocks esp32_pod_lid)

#	Measure	Feeds
14	Jack body X length (along the pod's X axis)	jack_body_l = 12 (PLACEHOLDER)
15	Jack body Y width	jack_body_w = 9.5 (PLACEHOLDER)
16	Jack body Z height below the lid, including screw terminals	jack_body_h = 11 (PLACEHOLDER)
17	Barrel outer diameter at the face (the part that goes through the lid)	barrel_jack_dia = 11.5

#	Measure	Feeds
18	Does it have a panel thread + nut? Thread dia and nut AF if so	decides hole vs cradle strategy
19	Polarity (multimeter: which terminal is centre-pin positive)	wiring, not CAD - but do it before power-on

ESP32 DevKit (blocks esp32_pod_shell, esp32_pod_params)

#	Measure	Feeds
20	Board length	devkit_length = 51.5
21	Board width	devkit_width = 28.0
22	Pin-row centre-to-centre pitch	M21 measured 25.6mm; hdr_row_pitch updated
23	Pin count per row	hdr_strip_len = 40
24	USB socket: width, height, and height of its underside above the board	usb_w = 10, usb_h = 7, usb_z = 12.5
25	Your USB cable's overmould width x height	usb_w, usb_h
26	Female header strip: body width and height with a board seated	hdr_strip_w = 2.7, hdr_strip_h = 8.5

ULN2003 driver module (blocks outer_box, mid_plate)

#	Measure	Feeds
27	Board length x width	elec_pocket 36 x 46 (outer_box)
28	Total height with wires soldered flat	elec_pocket_h = 16
29	Height and position of the tallest connector	mid_plate relief slot [3, ?23] cube ([17, 8])
30	Mounting-hole positions, if you intend to bolt it	mb_boss_positions = (+/-14, +/-19)

Magnets and switches

#	Measure	Feeds
31	Docking magnet diameter x thickness (re-confirm the new batch)	mag_dia = 8.4, mag_depth = 1.2
32	Homing magnet diameter x thickness (must be bought - 3x2 mm)	magnet_dia = 3.0, magnet_depth = 2.0
33	Tactile switch body W x D, total height including plunger, plunger travel	sw_body = 6.4, sw_depth = 5.5, cage depth 7.2, nav_shaft_len = 4.5
34	Switch lead spread and lead orientation	switch cage fin spacing

Pogo pins (not audited here - no dimensions exist in the CAD at all)

#	Measure	Feeds
35	Pogo carrier board width x height x thickness	pogo_carrier_w = 12, pogo_carrier_h = 10, pogo_carrier_d = 2 - all explicitly marked "measure real part!"
36	Pogo pin free length and compressed length	pogo_slot_cube ([6, 10, 8]), pogo_pad_recess = 1

11. BOTTOM LINE

Do not order resin yet, and do not attempt a full assembly.

Blocking failures that are certain from the source, independent of any measurement:

- 1 The hall sensor pocket **does not exist** - it is entirely inside the cam pocket at the same depth (§2a).
- 2 The cam homing-magnet pocket is a **through-hole that craters cam tracks 2 and 3** (§2c).
- 3 The motor's **right mounting screw hole opens into the spring cavity** - one-screw motor (§3a).

- 4 The motor's **left screw hole has a 0.35 mm wall** to the plate edge (§3b).
- 5 The **ESP32 board overruns the pod cavity by 1.75 mm** (§6a).
- 6 The **USB cutout is ~2 mm above the port** (§6c).
- 7 The **jack cradle overruns the pod cavity by 1.5 mm**, with placeholder dimensions (§7a).
- 8 The **cam z-stack does not close**: hub 2 mm too long, shaft ~6 mm too long for the open bore, putting all six dots permanently proud and the shaft into the dot-cluster risers (§4c).

What actually passes: the **dot insert / top plate** interface (§9), the **docking magnets** (§8), and the **ULN2003 in the pocket provided its wires are soldered flat** (§5).

The single highest-value next action remains the one in the handoff: **take the 36 measurements above**, starting with the motor, then re-derive the z-stack in one pass. Nothing downstream is trustworthy until the motor is a measured object rather than a datasheet.